

Unit 5, Lesson 6: Perimeter on the Coordinate Plane

Benchmarks

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.

Learning Targets

- ☐ I can find distances between ordered pairs.
- ☐ I can find the dimensions of a rectangle on a coordinate plane.
- ☐ I can find the perimeter of a rectangle on a coordinate plane.

5.6.1 Warm-Up: Self-Assessment

1. Complete each statement to describe your understanding.

- ☐ I know
- ☐ I need support with

how to plot rational number ordered pairs in all four quadrants.

- ☐ I know
- ☐ I need support with

how to find the perimeter and area of rectangles.

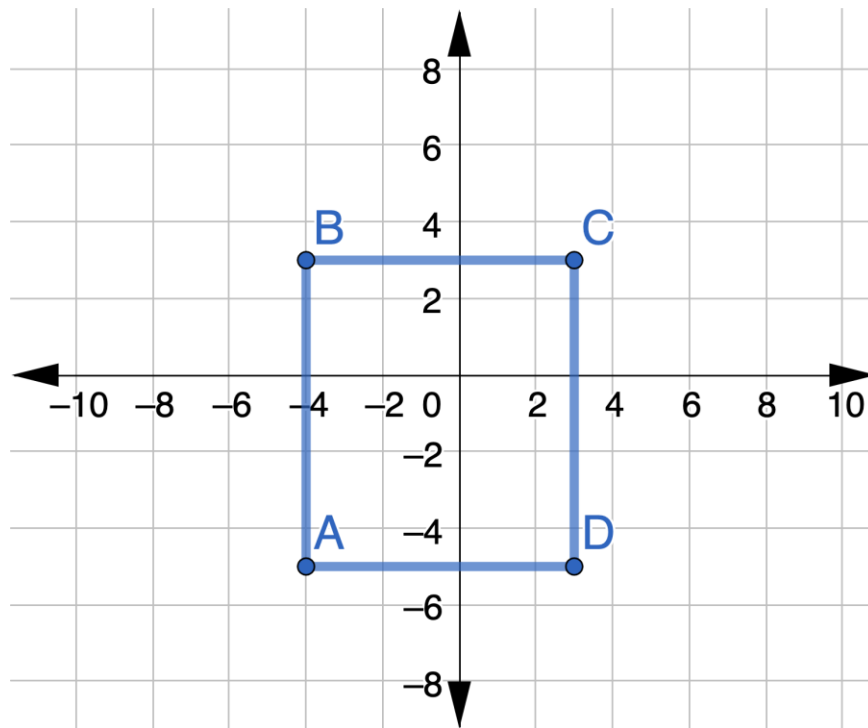
- ☐ I know
- ☐ I need support with

how to find the vertical or horizontal distance

between two ordered pairs that are located in the same or different quadrants.

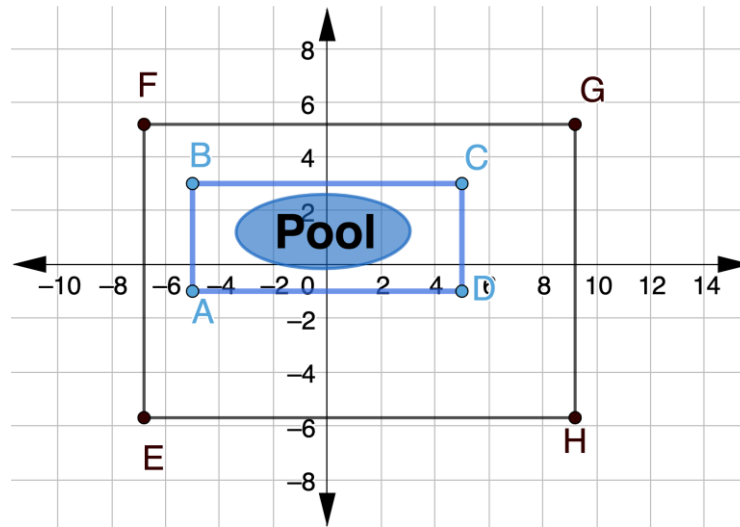
5.6.2 Guided Instruction: Dimensions and Perimeters of Rectangles

1. A rectangle has vertices at the points $A(-4, -5)$, $B(-4, 3)$, $C(3, 3)$, and $D(3, -5)$ as shown.



- The distance between points A and D is the length of the rectangle. The length of the rectangle is _____ units.
- What other two points on the rectangle are the same distance apart as points A and D ?
- Between which two points is the line segment that represents the width of the rectangle?
- Find the width of the rectangle.

2. Sonja's family installed a new pool. Sonja's dad wants to put up a fence, shown as a blue rectangle, around the pool for safety. The scale of the grid is represented in meters (m). He needs to know how much fencing he will need.



The rectangle that represents the fence has vertices at $A(-5, -1)$, $B(-5, 3)$, $C(5, 3)$, and $D(5, -1)$.

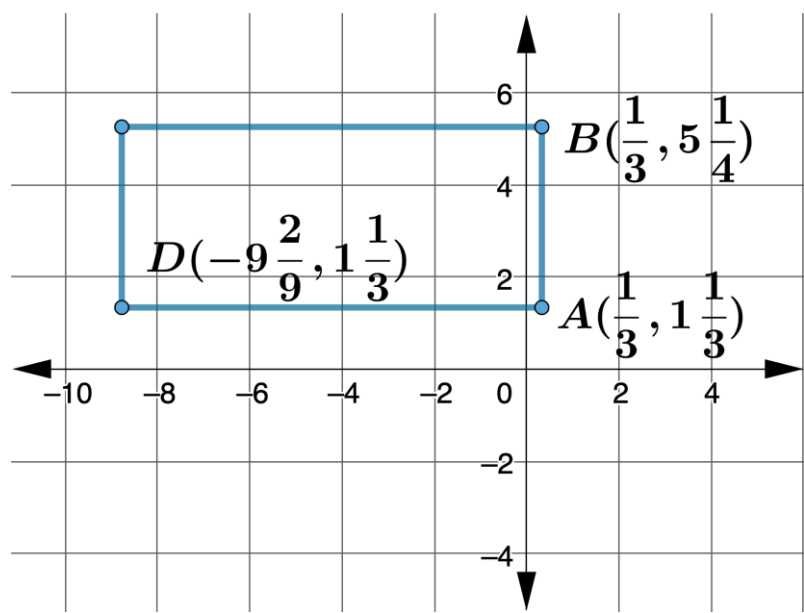
- What is the length of the line segment that connects points A and D ?
- What is the length of the line segment that connects points A and B ?
- Explain how to calculate the perimeter of the fence.
- What is the perimeter of the fence in meters?

5.6.3 Try It: Perimeter on the Coordinate Plane

1. Sonja's father plans to put another fence around the yard. The black rectangle on the coordinate grid in the previous component represents the second fence. The scale of the grid is represented in meters (m). The black rectangle has vertices at $E(-6.8, -5.7)$, $F(-6.8, 5.2)$, $G(9.2, 5.2)$, and $H(9.2, -5.7)$.
 - a. What is the length of the line segment that connects point E to point H ?
 - b. What is the length of line segment that connects point E to point F ?
 - c. Determine the perimeter of the fence around the yard.

5.6.4 Collaboration: Fix the Mistake

1. Vera and Conor are working on their homework together to find the perimeter of the rectangle, shown on the coordinate plane. The width of the rectangle is the segment between point *A* and point *B*.



Conor counts the units between the numbers, while Vera determines the dimensions by subtracting. Each student’s work and answers are shown.

Vera’s Thinking	Conor’s Thinking
Vera finds the dimension of the width by subtracting the <i>y</i>-coordinates. $5\frac{1}{4} - 1\frac{1}{3} =$ $\frac{21}{4} - \frac{4}{3} =$ $\frac{63}{12} - \frac{16}{12} =$ $\frac{47}{12}$ Vera says the width of the rectangle is $3\frac{11}{12}$ units.	Conor notices there is 1 full unit between point <i>A</i> and point <i>B</i> and two pieces that look like they would combine to make another whole unit. Conor says the width of the rectangle is 2 units.

Discuss each student's work with your partner. Explain any errors that either student made.

2. The length of the rectangle is the segment between point A and point D .

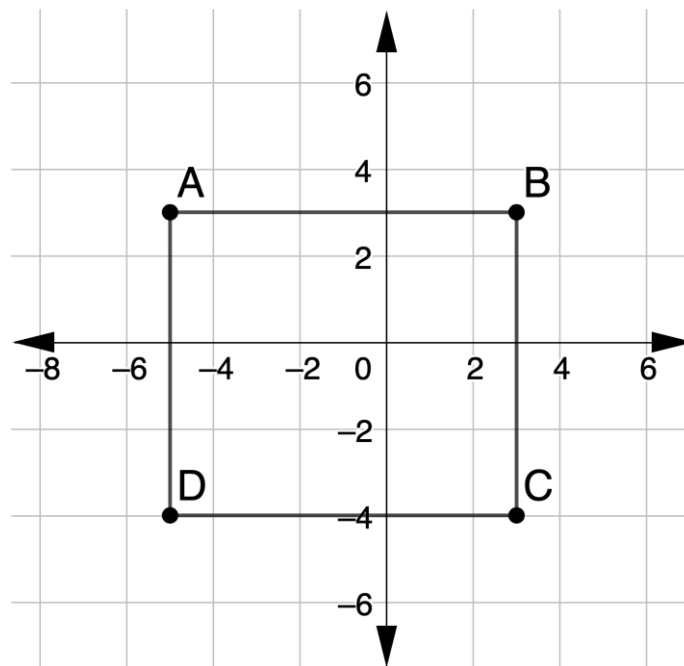
Vera finds the dimensions of the length by subtracting the x-coordinates. $9\frac{2}{9} - \frac{1}{3} =$ $\frac{83}{9} - \frac{3}{9} =$ $\frac{80}{9}$ Vera finds the length of the rectangle to be $8\frac{8}{9}$ units.	Conor counts the units between the numbers again. He counts 4 units between A and D, but then realizes each space on the grid represents 2 units, so he doubles this to 8. He then determines $\frac{2}{9}$ of a unit plus $\frac{1}{3}$ of another unit equals $\frac{5}{9}$. He adds this to the 8. Conor finds the length of the rectangle to be $8\frac{5}{9}$ units.
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Discuss each student's work with your partner. Explain any errors that either student made.

3. After collaborating with each other and understanding their mistakes, Vera and Conor each correctly find the perimeter of the rectangle. What perimeter did they find?

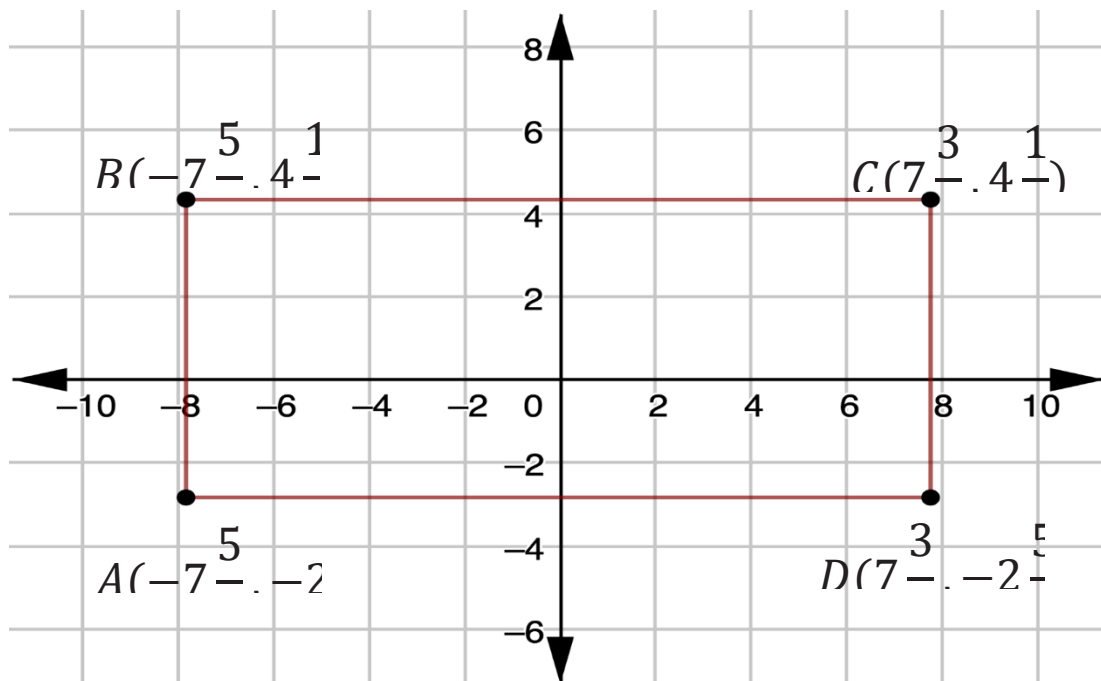
5.6.5 Your Turn!

1. The vertices of a rectangle are at points $A(-5, 3)$, $B(3, 3)$, $C(3, -4)$, and $D(-5, -4)$.



- What is the length, in units, of the rectangle?
- What is the width, in units, of the rectangle?
- What is the perimeter, in units, of the rectangle?

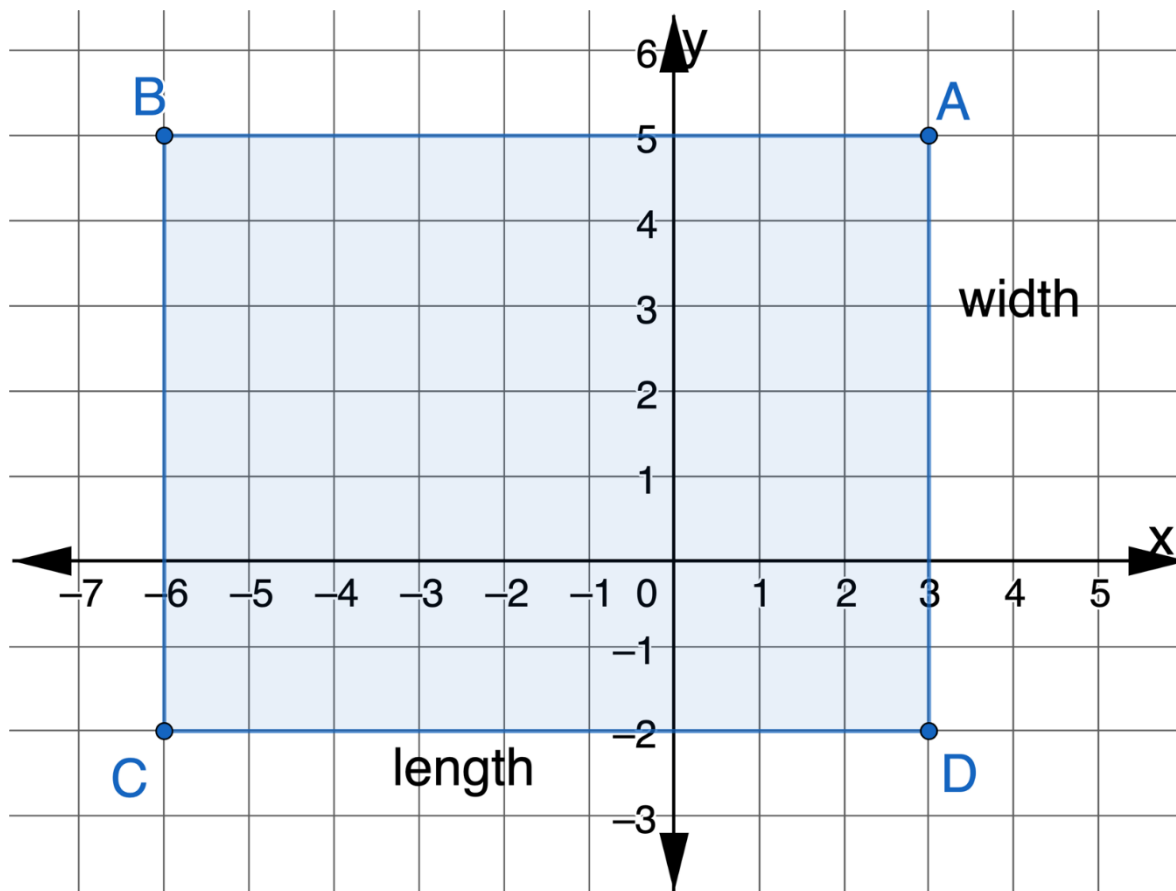
2. Titus wants to put a border around the outside of his garden.



If the scale of the grid is in feet (ft), how many feet of border will he need to buy to completely fence in the garden?

5.6.6 Wrap-Up: Ticket Out the Door

1. A rectangle is shown on the coordinate plane.



- What is the length, in units, of the rectangle?
- What is the width, in units, of the rectangle?
- What is the perimeter, in units, of the rectangle?